Midterm Organisations and Markets 2026

11 november 2026

1 [60 points] Entry and Competition

In a market for potatoes, demand is given by $D(p) = 630 - \frac{p}{2}$. If a firm produces potatoes, it has to pay a fixed cost of 100 and after that a cost $c(q) = 4q^2$ to produce q potatoes.

(a) Suppose there is a single price-taking firm. Calculate the equilibrium price and quantity.

(b) What are the Lerner index and the Herfindahl-Hirschman index for this equilibrium?

(c) Now suppose that firms can freely enter, but are still price taking. Calculate the equilibrium price and quantities (per firm and in total).

(d) What are the Lerner index and the Herfindahl-Hirschman index for this equilibrium?

(e) How do the two equilibria compare to each other for the consumers? For the firms? In terms of efficiency?

From now on assume firms are not price taking, but strategically competing by setting quantities.

(f) Suppose again that there is a single firm. Calculate the equilibrium price and quantity.

(g) What are the Lerner index and the Herfindahl-Hirschman index for this equilibrium?

(h) Now suppose there are two firms that simultaneously set a quantity to produce. Calculate the equilibrium price and quantity.

(i) What are the Lerner index and the Herfindahl-Hirschman index for this equilibrium?

Now suppose that firms can enter into this market at will (while still competing in quantity and still facing a fixed cost of 100).

(j) Suppose in this new situation the market demand increases to $D_{New}(p) = 1260 - p$. This will have two distinct effects on efficiency. Explain intuitively what they are and what their direction is. Show no calculations.

2 [40 points] Teams

Alice and Bob can work together to produce widgets, which they can sell at price 1. If Alice works A hours and Bob works B hours, they produce $Q(A, B) = 30A + 30B + AB$ widgets. Their cost of working is $C(A) = A^2$ and $C(B) = B^2$ respectively.

(a) Compute Alice's marginal productivity. What can you say about how Alice and Bob produce as a team? $\int_0^1$

(b) What would be the efficient production level?

(c) Suppose that Alice and Bob split the revenue equally and simultaneously choose how many hours to work. What will be the resulting equilibrium hours worked and equilibrium output?

(d) Now suppose Chrysostomos offers a prize P to Alice and Bob. Only one of them can get the prize and the probability that Alice will get it is $\frac{A}{A+B}$ and the probability that Bob will get it is $\frac{B}{A+B}$. Chrysostomos gets to keep and sell whatever Alice and Bob produce. How large does Chrysostomos have to set P in order to induce the efficient production level?